Brunda Marpadaga

Embedded Firmware Engineer | C/C++ | FreeRTOS | Embedded Linux | Board Bring-up
[redacted] | brundamarpadaga@gmail.com | LinkedIn | GitHub | Portfolio

Objective

Full-time Embedded Firmware Engineer role writing C/C++ drivers and RTOS or embedded Linux software for microcontroller- and SoC-based products, owning board bring-up and hardware/software debug from datasheet to working system, and growing depth on the MCU platforms the team ships.

Education

Portland State University — Portland, OR
M.S. Electrical and Computer Engineering – ***GPA 3.85/4*** — *Graduated Dec 2025*
- **Coursework:** Embedded Systems Programming, Microprocessor System Design, Computer Architecture

Technical Skills

Languages: C, C++, SystemVerilog, MATLAB
Firmware & OS: FreeRTOS, bare-metal, embedded Linux (PetaLinux/Yocto, device tree)
Interfaces: I2C, SPI, UART, GPIO, AXI, Ethernet, TCP/IP, MQTT
Hardware: STM32, ESP8266/Arduino, Xilinx Zynq-7000 SoC (Zybo Z7-20), Artix-7 (Nexys A7-100T), Veloce
Debug & Tools: JTAG, J-Link/Segger Ozone, GDB, STM32CubeIDE, Vitis, CMake, Git, oscilloscope, logic analyzer

Projects

Closed-Loop Light Control: I2C Sensor Driver + FreeRTOS | *C, FreeRTOS, I2C, PWM, MicroBlaze*
- Wrote a register-level I2C driver for the TSL2561 light sensor from its datasheet (init, gain/timing, lux math).
- Structured the application as FreeRTOS tasks for sensing, PID control and PWM output to hold LED brightness at a setpoint; brought up the target as a MicroBlaze soft-core on a Nexys A7 (Artix-7 FPGA) and debugged over JTAG.

Embedded Linux I2C Drivers + MQTT/TLS Telemetry | *C, PetaLinux/Yocto, Device Tree, Zynq-7020*
- Wrote user-space I2C drivers from datasheets for a BME280 sensor and SSD1306 OLED; configured Makefile and BitBake recipe to cross-compile and deploy an MQTT/TLS telemetry client.
- Fixed a non-probing Xilinx I2C driver by analyzing the device tree, kernel log (`dmesg | grep`) and kernel config.
- Debugged TLS handshake failures when publishing over MQTT via Ethernet; extended the system from local-only display to sensor-to-cloud telemetry by reusing the existing Linux network stack.

Dual-Core AMP Firmware on ARM Cortex-A9 | *C, FreeRTOS, Bare-metal, Zynq-7020*
- Partitioned a dual-core Cortex-A9 for asymmetric multiprocessing: FreeRTOS on Core 0 controlling a tiled HLS matrix-multiply accelerator, bare-metal C on Core 1 as the software baseline, with a split DDR memory map, a separate Core 1 linker script, and a shared-memory command/status mailbox signalled with SEV.
- Wrote an inter-core UART spinlock with LDREX/STREX and DMB barriers; double-buffered inputs with FreeRTOS tasks; read CP15 PMU and PL310 L2 counters (up to 9.4× HW speedup).

Interrupt-Driven IoT Infant Monitor | *C++, ESP8266, Wi-Fi/HTTP*
- Sampled 3 sensors on an ESP8266 using hardware interrupts for low-latency capture; drove a servo actuator and pushed telemetry over Wi-Fi/HTTP to ThingSpeak cloud with IFTTT webhook alerts to a phone.

Experience

Android Developer (Open Source Contributor) — Mar 2026 – Present
Greenstand — *Remote, U.S.*
- Built mobile features in Kotlin/Compose for the open-source Treetracker app, collaborating via Git reviews and GitHub Actions CI pipelines.

Associate Software Engineer — Jul 2022 – Dec 2023
Prodapt — *Hyderabad, India*
- Built full-stack web applications in Java Spring Boot and MongoDB in an Agile team using Git and Docker.

Failure Analysis Intern — May 2019 – Nov 2019
Electronics Corporation of India Limited (ECIL) — *Hyderabad, India*
- Debugged production Smart Energy Meter boards using test jigs, multimeter and JTAG to verify PCB hardware and firmware functionality, performed schematic-driven root-cause analysis, and fed corrective actions back to the soldering team.